

MUHAMMAD RIZQAN NUR ADJIE

AI / ML Engineer · Computer Vision · Frontend Developer

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PROFESSIONAL SUMMARY

Computer Science master's graduate (Tamkang University, 2026) with hands-on experience across computer vision, machine learning, and frontend development. Skilled in building and evaluating YOLO-based object detection models and tabular ML pipelines, and in high-end, animation-driven web with React, GSAP, and Three.js. Proficient in Python, PyTorch, and Ultralytics YOLO. Open to AI / ML, Computer Vision, and Frontend Engineering roles.

TECHNICAL SKILLS

Programming: Python, JavaScript, SQL, HTML, CSS

ML & Computer Vision: PyTorch, Ultralytics YOLO, Object Detection, scikit-learn, Transformers, Model Evaluation

Data Analysis: Pandas, NumPy, XGBoost, LightGBM, CatBoost, TabPFN, Exploratory Data Analysis

Frontend: React.js, Next.js, Tailwind CSS, GSAP, Three.js, Responsive Web Design

Backend & Tools: Node.js, FastAPI, Git, Figma, Roboflow

EDUCATION

M.Sc. Computer Science & Information Engineering | Tamkang University, Taiwan Sep 2024 – Jun 2026

B.Eng. Software Engineering | Telkom University, Indonesia Sep 2019 – Sep 2023

SELECTED PROJECTS

Tomato Maturity Detection Using YOLOv11 and Swin Transformer | Master's Thesis, Tamkang University

- Developed a YOLOv11-based pipeline to classify tomato maturity into green, orange, and red stages, integrating a Swin Transformer backbone to strengthen feature extraction under occlusion, scale variation, and lighting changes.
- Applied Multi-Scale SPPF refinement and Weighted Boxes Fusion to improve detection consistency across dense tomato clusters.
- Evaluated a 1,051-image dataset using mAP, precision, recall, and FPS, achieving 0.824 mAP@0.5 and 0.499 mAP@0.5:0.95.

Tourist Behavior Prediction Using Machine Learning | Research Project / Conference Paper Draft

- Built ML models to predict length of stay, total spending, satisfaction, and itinerary preference from 804 survey responses.
- Applied leakage-free preprocessing by removing post-experience variables and using pre-trip features only.
- Benchmarked 9 models (Logistic Regression, SVM, Random Forest, XGBoost, CatBoost, LightGBM, TabNet, FT-Transformer, TabPFN) with 5-fold stratified cross-validation; TabPFN v3 performed best on most targets.

Interactive Web Showcases | Personal / Creative Frontend Projects

- Built cinematic, scroll-driven web experiences including a Porsche 911 GT3 RS canvas image-sequence showcase and a luxury resort site with WebGL depth-parallax.
- Used vanilla JS with Vite, React, GSAP, Lenis, and Three.js to deliver smooth scroll-scrubbing, custom cursors, and responsive layouts.

RELEVANT EXPERIENCE

Frontend Developer | Capstone Cafeasy, Bandung, Indonesia Sep 2022 – Sep 2023

- Implemented frontend features and contributed to the architecture and design of web interface components.
- Supported performance and usability improvements through frontend maintenance and cross-team collaboration.

Data Analyst (FGA) | Digital Talent Scholarship, Bandung, Indonesia Sep 2022 – Nov 2022

- Completed data analyst training covering SQL, data cleaning, exploratory data analysis, and visualization through labs and project-based assessments.

Student Intern | PT Batulicin Enam Sembilan Group, Jakarta, Indonesia Jun 2022 – Aug 2022

- Migrated and analyzed company data for insight extraction, and redesigned digital interface layouts in Figma to improve usability.

AWARDS & SCHOLARSHIPS

- Yu-Lien Scholarship:** Awarded NT\$100,000 by Tamkang University for outstanding entrance performance (2024–2025).
- Graduate Scholarship:** Received Tamkang University postgraduate scholarship, second semester of AY 113 (2025).

LANGUAGES

Indonesian (Native) · English (Professional Working Proficiency)